

Prime Divisibility, Geometric Resonance, and Higher-Dimensional Projection

Penrose Tilings as a Lattice-Projection Case in the Hyperdimensional Ladder

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Abstract

Penrose tilings provide a precise mathematical case in which simple local rules, algebraic number fields, higher-dimensional projection, and global nonperiodic order become visible in one structure. They do not merely decorate a plane with attractive polygons. They show how a registry of measurement - here a lattice, a projection rule, an acceptance window, and a local matching system - can generate a stable, shareable, nonperiodic geometry whose order exceeds ordinary translational periodicity.

This paper presents Penrose tilings as a concrete cross-section of the broader Lambda / TUFT / NSAF hyperdimensional ladder. The core claim is not that one construction is “native” while the others are derivative. No mathematical map arrives from the territory as an unmediated object. Cut-and-project schemes, de Bruijn multigrids, substitution matrices, cyclotomic coordinates, toroidal phase representations, and Hopf-style bundle descriptions all impose rules of meaning. Each selects invariants by choosing an axiomatic process, a coordinate registry, and an equivalence relation.

Within that posture, the Penrose case remains exceptionally rigorous. Fivefold order arises from the fifth roots of unity and the golden field. Inflation acts through the algebraic unit

$$\phi = \frac{1 + \sqrt{5}}{2},$$

with tile-count growth governed by a Perron-Frobenius eigenvalue ϕ^2 . The cut-and-project method realizes the tiling as a two-dimensional selection from a higher-dimensional lattice. The de Bruijn pentagrid supplies an equivalent coordinate registry. The tiling hull, diffraction module, and substitution action carry spectral and dynamical invariants. Prime factorization in $\mathbb{Z}[\phi]$ and cyclotomic fields organizes the arithmetic environment in which these rotations, scalings, and resonances occur.

The result is a disciplined formulation of the larger thesis: simple causal or constructive rules, iterated under number-theoretic constraints and projected through selected measurement cuts, generate complex resonant structures. Penrose tilings give a canonical case where this statement can be stated without mysticism and without reducing geometry to an allegedly frame-free “physical” object. The tiling is a named form: a stable, constrained, axiomatized registry of higher-dimensional relations.

1. Claim posture and ontological frame

This paper uses the word “construction” in a strict sense. A construction is an imposed formal process that selects, relates, and stabilizes a set of distinctions. A construction may reveal powerful invariants, but it does not escape framing. It supplies the rules by which something becomes comparable, nameable, and shareable.

Thus, the paper does not seek a privileged “native construction” of Penrose tilings. The cut-and-project method, the de Bruijn pentagrid, substitution inflation, cyclotomic algebra, and Hopf-style phase lifting all function as imposed mathematical registries. Each one makes certain relations exact. Each one hides or compresses others. Their value lies not in being unframed, but in preserving structure across transformations.

The guiding principle is:

stable form = constraint+rule of selection+equivalence relation+persistence under transformation.

Penrose tilings satisfy this principle with unusual clarity. Their local matching rules constrain adjacency. Their algebraic units constrain scaling. Their projection windows constrain selection.

Their spectral module constrains long-range order. Their aperiodicity prevents reduction to a simple translational cell.

2. Penrose tilings as the working exemplar

A Penrose tiling is a nonperiodic tiling of the plane with finite local complexity and global long-range order. Standard versions use either kites and darts or thick and thin rhombi. The rhombus version uses angles associated with fifth roots of unity: 36° , 72° , 108° , and 144° . The tilings display fivefold rotational order locally and statistically, but no nonzero translational period.

The essential features are:

1. **Aperiodicity**: no translational period repeats the whole tiling.
2. **Finite local complexity**: only finitely many local patch types occur up to translation for any bounded radius.
3. **Repetitivity**: every finite patch reappears throughout the tiling with bounded gaps.
4. **Inflation symmetry**: the tiling can be scaled by ϕ and decomposed into tiles of the original type.
5. **Higher-dimensional representation**: the tiling can be produced by projection from a higher-dimensional lattice.
6. **Algebraic control**: the relevant rotations and scalings live in fields generated by $\sqrt{5}$ and $\zeta_5 = e^{2\pi i/5}$.

These features make Penrose tilings an ideal case for the hyperdimensional ladder: a visibly two-dimensional object whose order is more naturally expressed by a higher-dimensional arithmetic and projection registry.

3. The golden field and fifth cyclotomic field

The golden ratio is

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \phi^2 = \phi + 1, \quad \phi^{-1} = \phi - 1.$$

The field generated by ϕ is

$$\mathbb{Q}(\phi) = \mathbb{Q}(\sqrt{5}).$$

Its ring of integers is

$$\mathbb{Z}[\phi] = \{a + b\phi : a, b \in \mathbb{Z}\}.$$

The fifth roots of unity generate the cyclotomic field

$$\mathbb{Q}(\zeta_5), \quad \zeta_5 = e^{2\pi i/5},$$

whose minimal polynomial over $\mathbb{Q}$ is

$$\Phi_5(x) = x^4 + x^3 + x^2 + x + 1.$$

The real subfield of $\mathbb{Q}(\zeta_5)$ is $\mathbb{Q}(\sqrt{5})$, since

$$\zeta_5 + \zeta_5^{-1} = 2 \cos\left(\frac{2\pi}{5}\right) = \frac{\sqrt{5}-1}{2} = \phi - 1.$$

This algebraic fact ties the fivefold angular registry directly to the square-root ladder rung at $\sqrt{5}$.

3.1 Prime factorization in the golden field

The discriminant of $\mathbb{Q}(\sqrt{5})$ is 5. Hence the prime 5 ramifies. For rational primes $p \neq 5$, splitting in $\mathbb{Q}(\sqrt{5})$ is controlled by whether 5 is a quadratic residue modulo p . Equivalently:

$$p \equiv \pm 1 \pmod{5} \Rightarrow p \text{ splits,}$$

$$p \equiv \pm 2 \pmod{5} \Rightarrow p \text{ remains inert,}$$

and

5 ramifies.

This does not mean that every tile adjacency is directly caused by a prime. It means the arithmetic module supporting the tiling's rotations, scalings, and diffraction data lives in a number field whose divisibility structure is prime-sensitive. Prime divisibility is therefore not incidental to the algebraic registry. It participates in the permitted factorizations of lengths, phases, modules, and recurrence scales.

4. Cut-and-project construction from a higher-dimensional lattice

Let $L = \mathbb{Z}^5$. Choose a two-dimensional subspace $E_{\parallel} \subset \mathbb{R}^5$ associated with fivefold symmetry, and let $E_{\perp}$ be its orthogonal complement. Let

$$\pi_{\parallel} : \mathbb{R}^5 \rightarrow E_{\parallel}, \quad \pi_{\perp} : \mathbb{R}^5 \rightarrow E_{\perp}$$

be the corresponding projections.

Given a bounded acceptance window

$$W \subset E_{\perp},$$

the projected point set is

$$\Lambda(W) = \{\pi_{\parallel}(n) : n \in \mathbb{Z}^5, \pi_{\perp}(n) \in W\}.$$

Under the standard irrational orientation, the projected full lattice may be dense in $E_{\parallel}$, but the window selection produces a Delone set: it is uniformly discrete and relatively dense. The tiling is then recovered by local adjacency or dual-cell rules applied to this projected set.

This is the first rigorous expression of the ladder claim. A two-dimensional nonperiodic geometry is obtained by selecting a finite-thickness slice through a higher-dimensional periodic lattice. The nonperiodicity arises because the selected plane is irrational with respect to the lattice. A translational period in the projected tiling would lift to a compatible rational relation in the higher-dimensional lattice, but the chosen irrational orientation prevents such closure.

5. de Bruijn pentagrid coordinates

The de Bruijn pentagrid gives an equivalent imposed registry. Define five unit vectors in the plane:

$$v_j = \left(\cos \frac{2\pi j}{5}, \sin \frac{2\pi j}{5} \right), \quad j = 0, 1, 2, 3, 4.$$

For each j , take a family of parallel lines

$$x \cdot v_j + \gamma_j = k, \quad k \in \mathbb{Z},$$

where γ_j is an offset parameter. These five line families form a pentagrid.

For a point $x \in \mathbb{R}^2$, define integer coordinates

$$K_j(x) = \lceil x \cdot v_j + \gamma_j \rceil.$$

The vector

$$K(x) = (K_0(x), K_1(x), K_2(x), K_3(x), K_4(x)) \in \mathbb{Z}^5$$

places the planar point into a five-coordinate registry. The Penrose tiling is obtained as the dual combinatorial structure generated by this multigrid. In this form, the visible two-dimensional tiling already carries five directional coordinates. The “plane” has not escaped the higher-dimensional relation; it records it.

The pentagrid and cut-and-project constructions express the same structural fact in two languages:

$$\text{five-coordinate lattice registry} \longrightarrow \text{two-dimensional aperiodic order}.$$

6. Substitution and inflation dynamics

Penrose tilings admit inflation and deflation. Linear lengths scale by ϕ . Tile counts scale by a matrix whose dominant eigenvalue is ϕ^2 , reflecting area growth.

Using a two-tile count convention with large/thick tiles A_L and small/thin tiles A_S , one standard substitution matrix is

$$M = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

This acts on count vectors by

$$\begin{pmatrix} A'_L \\ A'_S \end{pmatrix} = M \begin{pmatrix} A_L \\ A_S \end{pmatrix}.$$

The characteristic polynomial is

$$\chi_M(\lambda) = \lambda^2 - 3\lambda + 1.$$

Thus the eigenvalues are

$$\lambda_1 = \frac{3 + \sqrt{5}}{2} = \phi^2, \quad \lambda_2 = \frac{3 - \sqrt{5}}{2} = \phi^{-2}.$$

The Perron-Frobenius eigenvector may be taken as

$$\begin{pmatrix} \phi \\ 1 \end{pmatrix},$$

since

$$M \begin{pmatrix} \phi \\ 1 \end{pmatrix} = \phi^2 \begin{pmatrix} \phi \\ 1 \end{pmatrix}.$$

Therefore the asymptotic large-to-small tile ratio is

$$A_L : A_S = \phi : 1.$$

The powers of M are expressed by Fibonacci numbers. With $F_1 = 1$, $F_2 = 1$, $F_3 = 2$, and so on,

$$M^n = \begin{pmatrix} F_{2n+1} & F_{2n} \\ F_{2n} & F_{2n-1} \end{pmatrix}.$$

For example:

$$M^1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},$$

$$M^2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix},$$

$$M^3 = \begin{pmatrix} 13 & 8 \\ 8 & 5 \end{pmatrix}.$$

This is the second rigorous expression of the ladder claim. The same algebraic unit ϕ that generates fivefold scaling also controls the growth eigenvalue of the inflation registry.

7. Matching rules as local resonance constraints

Matching rules are local constraints placed on tile edges, vertices, colors, arcs, orientations, or decorations. They prevent arbitrary periodic arrangements and enforce global aperiodic order.

In the Lambda / TUFT / NSAF language, a matching rule functions as a local resonance constraint. It says which adjacencies may persist under the registry. It does not merely decorate the tiles; it carries global order through local enforcement.

Formally, let $\mathcal{T}$ be the finite set of prototiles and let $\mathcal{A}$ be the set of admissible adjacency relations. A tiling T is admitted when every local adjacency in T belongs to $\mathcal{A}$. The set of all admitted tilings forms a tiling space or hull under translation. Penrose matching rules define a hull with finite local complexity, repetitivity, nonperiodicity, and inflation structure.

This local-to-global constraint is the geometric analogue of a resonance condition:

$$\text{allowed local relation} \longrightarrow \text{global persistence under iteration.}$$

8. Spectral and diffraction structure

Penrose tilings possess long-range order even without translational periodicity. Their diffraction spectrum is pure point: the Fourier transform of the autocorrelation measure consists of Bragg peaks supported on a dense Fourier module generated by the projected higher-dimensional lattice.

This fact gives a precise meaning to “resonance.” The tiling does not repeat periodically in physical coordinates, but its spectral registry contains sharp ordered frequencies. The pattern is nonperiodic in translation space and ordered in reciprocal space.

Let $\Lambda(W)$ be the cut-and-project point set. Its Fourier module arises from the dual lattice projected into the physical reciprocal space. Intensities depend on the Fourier transform of the acceptance window W . Thus the projection window, lattice, and algebraic orientation jointly determine the spectral order.

This is not an analogy imposed after the fact. It is the spectral counterpart of the same cut-and-project construction.

9. Tiling hulls and dynamical zeta data

Let Ω_T denote the hull of a Penrose tiling T : the closure of all translates of T in the local matching topology. The plane acts on Ω_T by translation. Inflation acts as an additional dynamical operation when the substitution structure is included.

Different zeta functions may be attached to different imposed registries:

1. **Symbolic substitution registry:** periodic words or supertile recurrences.
2. **Inflation registry:** fixed tilings or periodic points under substitution powers.
3. **Cohomological registry:** Lefschetz-type traces on approximant complexes.
4. **Spectral registry:** diffraction and Fourier-module data.

The Artin-Mazur form for a map f is

$$\zeta_f(z) = \exp \left(\sum_{n=1}^{\infty} \frac{p_n}{n} z^n \right),$$

where p_n counts fixed points of f^n in the chosen dynamical registry. The substitution matrix M does not by itself define all such counts, but it supplies the leading inflation growth scale through ϕ^2 , and its traces encode the linearized tile-count recurrence:

$$\text{tr}(M^n) = F_{2n+1} + F_{2n-1} = L_{2n},$$

where L_n denotes the Lucas numbers.

Thus the zeta discussion should not collapse all dynamical invariants into one formula. The correct statement is stronger and cleaner: Penrose tilings support several compatible invariant registries, all governed by the same inflation and algebraic field data.

10. Prime divisibility as arithmetic resonance

Prime divisibility enters at three levels.

10.1 Field level

The fields $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\zeta_5)$ determine the arithmetic environment for fivefold symmetry. Rational primes split, ramify, or remain inert according to their behavior in these fields. This controls factorization in the coordinate and scaling rings.

10.2 Scaling level

The unit $\phi \in \mathbb{Z}[\phi]$ satisfies

$$N(\phi) = \phi(1 - \phi) = -1,$$

so ϕ is invertible in the ring of integers. Inflation and deflation are therefore algebraically reversible at the unit level, even though a particular finite patch may gain or lose resolution under a chosen representation.

10.3 Recurrence level

In cyclic or symbolic systems, primitive cycles and co-prime periods delay total realignment. If two periods m and n satisfy

$$\gcd(m, n) = 1,$$

then

$$\text{lcm}(m, n) = mn.$$

This maximizes the combined return time relative to the two cycle lengths. Prime periods are special cases of irreducible recurrence. This provides the precise mathematical sense in which “prime resonance” can be used without overclaiming: primes and primitive factors mark irreducibility, delayed coincidence, and factorization constraints within cyclic registries.

11. Hopf, toroidal, and phase-bundle interpretation

The Hopf fibration is

$$S^1 \longrightarrow S^{2n+1} \longrightarrow \mathbb{CP}^n.$$

It supplies a general phase-bundle registry: circle fibers over a projective base. In the hyperdimensional ladder, this is not treated as a privileged native map. It is another imposed construction, valuable because it makes phase, fiber, projection, and base relations explicit.

Penrose tilings already contain angular and phase-like structure through fifth roots of unity, multi-grid offsets, acceptance windows, and torus-parametrized hull data. The Hopf-style interpretation enters when those phase relations are lifted into a bundle language:

$$\text{local phase / fiber} \longrightarrow \text{projected base relation} \longrightarrow \text{registered tiling cross-section}.$$

This aligns the Penrose case with the larger TUFT / NSAF vocabulary. The tiling becomes a two-dimensional cross-section of higher-dimensional ordered relations. Its polygonal matching rules and inflation steps appear as visible residues of deeper phase, projection, and arithmetic constraints.

12. Relation to strange attractors and cymatic structures

Penrose tilings should not be identified carelessly with Lorenz-type strange attractors. They belong to a different mathematical class. Yet they share a structural pattern:

simple rule + iteration + constraint $\longrightarrow$ complex global form.

For strange attractors, deterministic equations generate bounded chaotic recurrence in phase space. For Penrose tilings, local matching, substitution, and projection generate a nonperiodic but ordered tiling hull. For cymatic and Chladni patterns, wave equations and boundary conditions generate nodal forms. These are not the same object. They are comparable meta-structures because each arises from an imposed rule system that stabilizes complexity through constraint.

In this sense, the analogy is structural rather than taxonomic. Lorenz dynamics, Penrose inflation, and cymatic excitation all demonstrate that simple causal or constructive rules can yield forms whose global complexity far exceeds the apparent simplicity of the local rule.

13. The hyperdimensional ladder

The hyperdimensional ladder begins from constructible and algebraic extensions:

$$\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt{3}, \sqrt{5}, \dots)$$

with special rungs associated to algebraic units and cyclotomic symmetries. The Penrose case occupies the $\sqrt{5}$ and fifth-cyclotomic rung. It uses:

$$\sqrt{5}, \quad \phi = \frac{1 + \sqrt{5}}{2}, \quad \zeta_5 = e^{2\pi i/5}, \quad \mathbb{Z}[\phi].$$

The tiling is therefore a concrete two-dimensional slice through a higher-dimensional algebraic-topological registry. It shows how a local plane can carry a nonlocal arithmetic order.

Within the traversal-engine vocabulary, the pilot does not impose arbitrary decoration on an empty plane. The pilot selects offsets, scales, projections, and phase relations inside an already constrained construction. Navigating the tiling means navigating a registry of algebraic possibilities.

14. Consolidated theorem: Penrose tiling as ladder cross-section

Theorem

A Penrose tiling can be represented as a two-dimensional registry of a higher-dimensional lattice selected by an irrational projection, constrained by a bounded acceptance window, ordered by fifth-cyclotomic directions, and recursively scaled by the golden unit ϕ . Its local matching rules, inflation dynamics, spectral module, and tile frequencies are compatible expressions of that same algebraic-projection structure.

Proof sketch

1. The fifth roots of unity define five angular directions and generate the cyclotomic field $\mathbb{Q}(\zeta_5)$.
2. The real subfield is $\mathbb{Q}(\sqrt{5})$, where the golden unit ϕ satisfies $\phi^2 = \phi + 1$.
3. The cut-and-project construction selects points from $\mathbb{Z}^5$ by projecting to a two-dimensional plane and filtering through a bounded internal window.

4. The irrational orientation prevents translational periodicity in the projected tiling.
5. The substitution matrix $M = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ has Perron-Frobenius eigenvalue ϕ^2 , giving inflation growth and tile-frequency convergence.
6. Matching rules enforce local compatibility so that the global aperiodic order persists.
7. The diffraction and hull invariants record the same projection and algebraic data in spectral and dynamical registries.

Therefore, the Penrose tiling functions as a rigorous ladder cross-section: a stable two-dimensional named form produced by higher-dimensional lattice projection, algebraic scaling, local rule enforcement, and spectral persistence. $\square$

15. Conclusion

Penrose tilings show how a finite set of local rules and algebraic constraints can generate infinite nonperiodic order. Their structure is not accidental. It follows from the interaction of fifth-cyclotomic symmetry, the golden field, higher-dimensional projection, substitution inflation, and local matching.

The larger Lambda / TUFT / NSAF reading does not require a privileged unframed construction. It treats every mathematical representation as an imposed registry. The strength of the Penrose case lies in the compatibility among its registries: cut-and-project, pentagrid, substitution, spectral, arithmetic, and phase-bundle descriptions all preserve the same core relations.

This is why Penrose tilings serve as a clean example of the hyperdimensional ladder. They show that what appears as a two-dimensional tiling can carry higher-dimensional arithmetic order. They also show how simple constructive rules, when iterated under number-theoretic constraint, produce resonant complexity without randomness and without ordinary periodic repetition.

The lesson generalizes without mystical force: named form arises where constraint, rule, projection, and persistence become stable enough to share.

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